Staff data management handbook

PRIOR + PTNRS



DD Month YYYY

Issued by Prior + Partners
Digital Innovation Team

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Part I
General guidelines

01
Purpose & scope

This handbook provides practical guidance for all Prior + Partners staff who work with spatial and non-spatial data. It covers how to connect to the company geodatabase, navigate available datasets, and manage data within your projects.

01.1
Why does this matter?

— Saves time by making data easy to find and understand
— Prevents data loss and duplication
— Ensures consistent quality across all projects
— Enables effective collaboration between teams
— Facilitates GIS data submission during later project stages by ensuring datasets are properly organised, named, and ready for handover

01.2
Who is this for?

— Project leads, analysts, designers, and all colleagues who use spatial data
— Anyone loading data into QGIS or ArcGIS for visualisation or analysis
— New starters setting up their database connection for the first time

01.3
What does this handbook cover?

— Understanding the company geodatabase and its structure
— Setting up your personal connection to the database
— Navigating and importing datasets into your project workspace
— Saving and naming data correctly during project work

02 Roles & responsibilities

Understanding who does what helps you work with data efficiently and know where to go for support.

02.1
Database administrator

Abhimanyu Acharya, Po Nien Chen, Sandra Wilmot

— Maintain the central geodatabase and all baseline datasets
— Manage database access, permissions, and security
— Process and quality-check datasets before they become available to projects

02.2
Digital innovation team

Members of DI team

— Set up project schemas and configure QGIS workspaces for new projects
— Bridge between the database and project-specific needs
— Support project teams with spatial data requests and analysis
— Import project-specific data into assigned schemas
— Conduct geospatial analysis (network, spatial, attribute-based)

02.3
Project teams

Project leads, analysts, designers, and all colleagues working with spatial data

— Load approved datasets into QGIS or ArcGIS from the geodatabase
— Apply symbology and produce visualisation outputs in P+P style
— Save and name project data following the standards in this handbook
— Flag data gaps or additional dataset requirements to the DI team

For any issues with accessing or using data in the geodatabase, contact Abhi, Po, and Sandra or DI team members.

Part II
About the Geodatabase

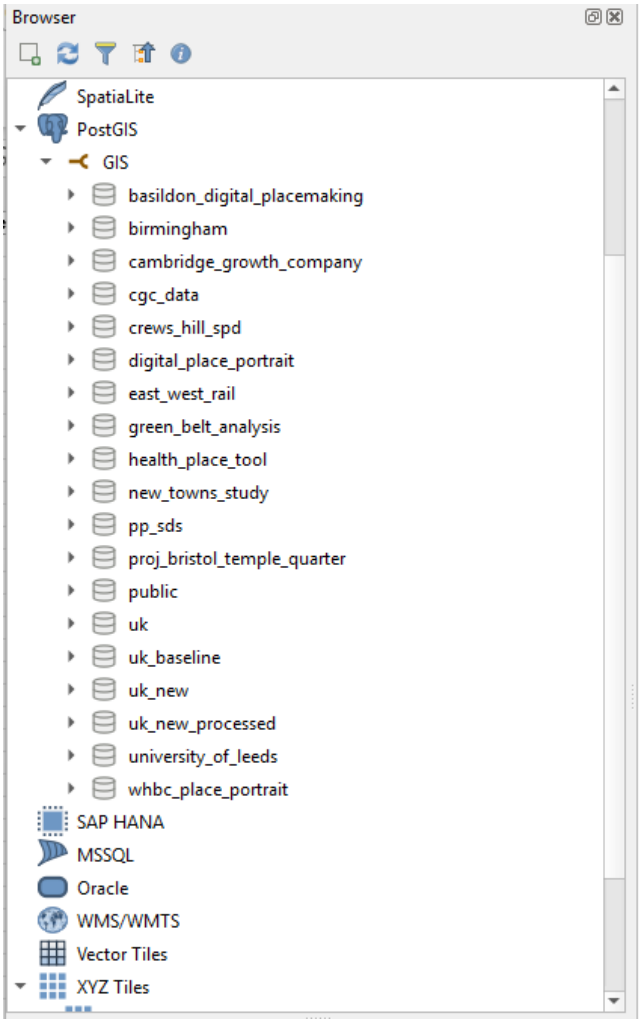
03
What is the Geodatabase?

The geodatabase is the company's central spatial data system. It stores all geographic datasets in one place, replacing scattered files across individual machines and project folders. This section explains what it is, why we use it, and how it's organised.

03.1

What you will see in QGIS

When you connect to the database in QGIS, the Browser Panel will list all available schemas under the PostGIS connection.



03.2

What is PostGIS?

PostGIS is a spatial database built on PostgreSQL — an industry-standard open-source database system. Think of it as a shared library for all of our geographic data. Instead of each person downloading and storing their own copy of a dataset, everyone connects to the same source.

You can access it through QGIS or ArcGIS, where layers from the database appear and behave just like any other data layer.

03.3

Why do we use it?

— Single source of truth: everyone works from the same approved datasets
— No more duplicate files across project folders and personal machines
— Baseline data is maintained and updated centrally by the DI team
— Project-specific data is kept separate from company-wide datasets
— Data is always available without needing to copy files or ask colleagues for links
— Data stays up to date: edits and updates sync to the database, so everyone's project workspace reflects the latest version. No more version conflicts between files

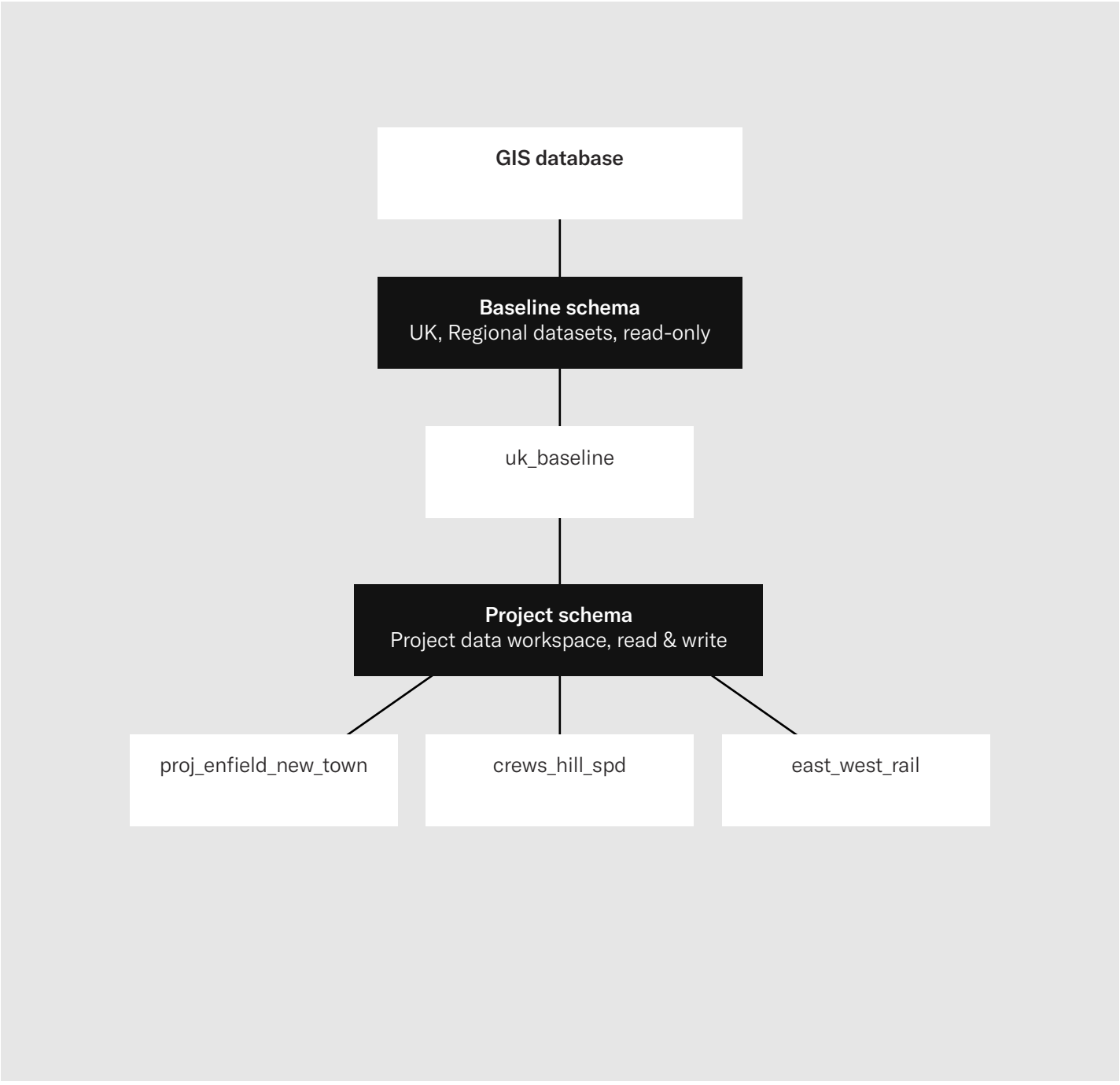
03.4

How is it organised?

The database is divided into schemas, which are separate containers that keep data organised by project and purpose.

— Baseline schema: UK-wide datasets maintained by the DI team covering themes such as transport, demographics, environment, and built form. This is the starting point for most project work. You have read-only access.

— Project schemas: dedicated spaces for each active project, set up by the DI team with baseline data filtered to the project area. You have write access here.



04 Datasets & Themes

All datasets in the geodatabase are organised by theme. Understanding these themes helps you quickly find the right data when browsing the database in QGIS.

04.1

How to read a layer name

<theme>_<region>_<source>_<layername>_<time>

— theme: short code indicating the data category (see table in p.19)

— <region> (if applicable): uk = United Kingdom, ew = England & Wales. An optional region code (uk, ew) occasionally follows the theme for a few datasets; most layers omit it.

— source: where the data comes from (os = Ordnance Survey, osm = OpenStreetMap, ons = ONS, dfe = Department for Education)

— layername: descriptive content identifier

— time: dataset period or last update (e.g. apr2025, 2023)

Example: hth_nhs_facility_2025

→ hth = Health & Wellbeing
→ nhs = NHS source
→ facility = the content description
→ 2025 = dataset year

04.2

Where to find data

— The uk_baseline catalogue: prior-partners.github.io/PP_uk_baseline
A webpage for every layer: what it is, where it came from, its licence, a preview map, and every column explained.

— Excel: P:\0_Practice\10_Data management resources\04_Spreadsheets\uk_baseline_catalogue.xlsx

— The Dashboard (Beta). An interactive map of every theme on your own machine — explore the data visually before you open QGIS. Releasing soon.

04.3

What a layer page shows

Every layer in the catalogue has its own page — open it to understand the data before you load it. Each page shows:

— Description: what the layer is, and its grain, in plain English.

— Source, licence & docs: where it came from and how you may use it.

— Preview map: its coverage and styling, at a glance.

— Every column explained: a table of each field with its meaning and units.



04.4

Data themes

These are the theme codes currently in use — 139 layers across 12 active themes. SAT and RST are reserved for the imagery and raster data we plan to add

	THEME	DATASET
ADM	**Administrative & Planning Boundaries**	Local authority, wards, output areas, built-up areas, postcode boundaries, planning zones, workplace zones
BLT	**Built Environment & Land Use**	Building footprints, land use classifications, planning applications, retail centres, points of interest
COM	**Community & Social Infrastructure**	Sports facilities, libraries, community centres, places of worship, cultural venues
DEM	**Demographics & Socio-Economic**	Census data (age, ethnicity, religion), deprivation indices, income data, household, travel-to-work data
ECN	**Economy & Employment**	Employment by sector, business registers, economic output, unemployment rates
EDU	**Education**	Schools database, Ofsted ratings, university locations
ENV	**Environment & Natural Assets**	Protected sites, woodland, green spaces, flood zones, air quality, noise monitoring, radiation monitoring
HER	**Heritage & Historical Assets**	Listed buildings, conservation areas, historic parks, archaeological sites
HOU	**Housing & Living Conditions**	House prices, housing tenure, property types, energy efficiency ratings, overcrowding metrics, rental costs
HTH	**Health & Wellbeing**	GP surgeries, hospitals, health inequalities, disease prevalence, obesity
MOB	**Transport & Mobility**	Road networks, railway lines, public transport routes, cycling infrastructure, stations, paths, trails
UTL	**Infrastructure & Utilities**	Electricity networks, gas pipelines, water infrastructure, telecoms masts, waste facilities
SAT	**Satellite Imagery**	Sentinel imagery, Landsat, aerial photography, vegetation indices, land cover classification
RST	**Raster & Grid Datasets**	Digital terrain models, climate data, soil maps, flood risk maps, population density grids

Part III
Personal Setup

05
Database connection setup

This section walks you through connecting QGIS to the company geodatabase. This is a one-time setup — once configured, QGIS will remember your connection for future sessions.

05.1

Setting up the connection

01 Open QGIS.

02 In the Browser Panel (left side of the screen), locate PostGIS (identified by the elephant icon).

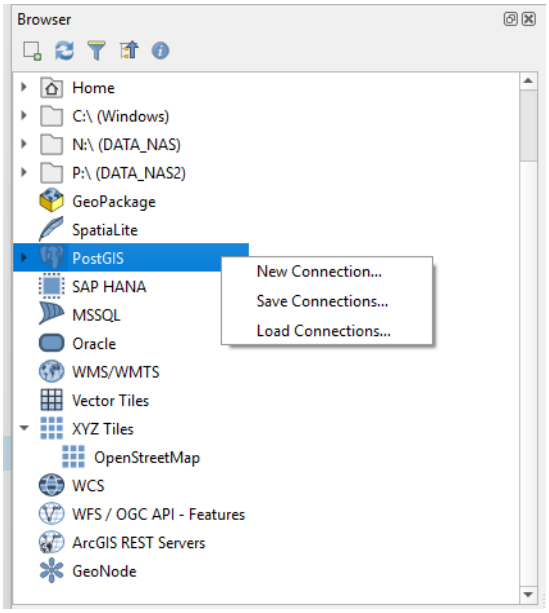
03 Right-click PostGIS and select New Connection.

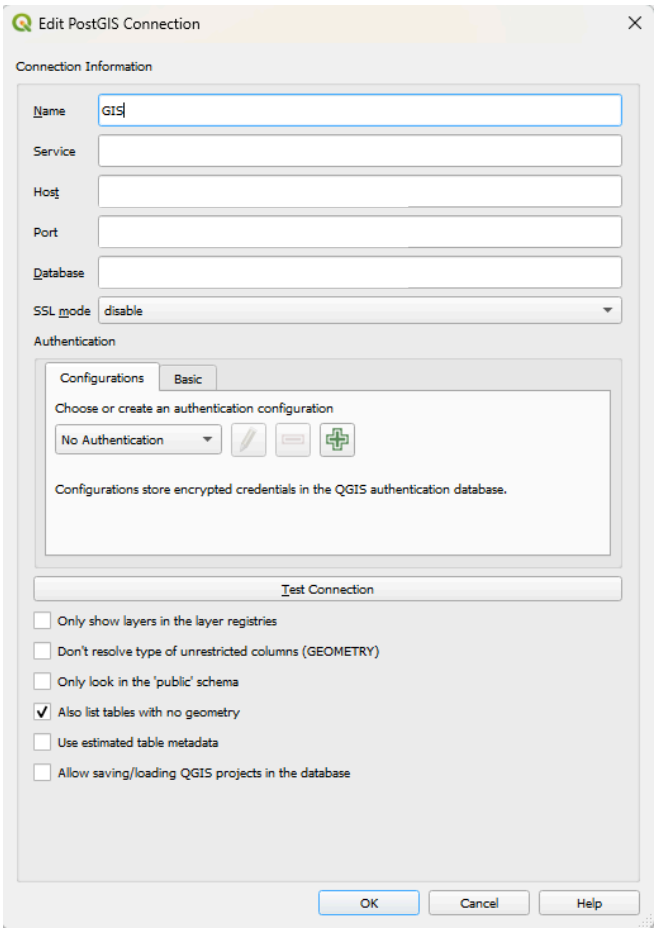
04 In the connection dialogue box, enter the following details:
— Name: provided by the DI team
— Service: Leave blank
— Host: provided by the DI team
— Port: provided by the DI team
— Database: provided by the DI team
— SSL Mode: Leave as “Disable”

05 Under the options at the bottom tick:
— Also list tables with no geometry

06 Click Test Connection.

07 If successful, click OK to save.





05.2

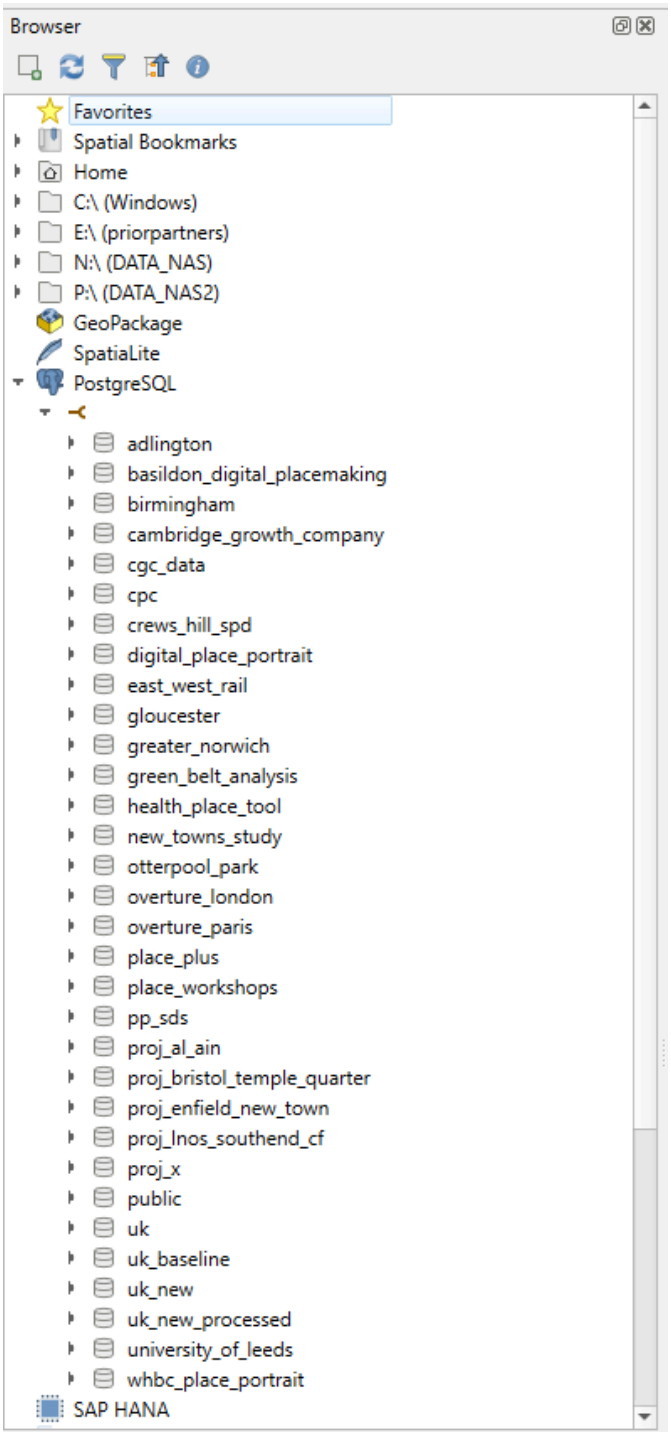
Verifying the connection

01 In the Browser Panel, expand the PostGIS section (click the arrow next to the elephant icon).

02 You should see your GIS connection. Expand it to view the available schemas (e.g. uk_baseline).

03 Expand to view all available schemas. You will see baseline schemas (e.g. uk_baseline), project schemas (e.g. proj_bristol_temple_quarter), and others.

If you can see the schema list, your connection is working.



05.3

Troubleshooting

— Connection failed: Check you are connected to the office server or VPN. The database is only accessible on the internal network.

— Incorrect credentials: Ensure the host, port, and database name are entered exactly as shown in 05.1. A single typo will prevent connection.

— Still not working: Contact Abhi, Po, or Sandra for assistance.

Part IV
Working on Projects

06
Project workspace

When a new project begins, the DI team prepares a ready-to-use QGIS workspace tailored to your project area. This section explains what's included, how to open it, and what you can and can't edit

06.1

What the DI team prepares for you

Before your project starts, the DI team sets up three things:

— Project schema: a dedicated space in the geodatabase for your project data (e.g. proj_bristol_temple_quarter). This is where you store project-specific datasets.

— Baseline layers filtered to your area: relevant UK-wide datasets clipped or filtered to your project boundary, pre-loaded into the workspace.

— QGIS project file (.qgz): a pre-configured workspace saved to the project server, containing the baseline layers, project boundary, and connection to your project schema.

The QGIS project file is stored at:
N:\00_XX_PROJECT\WORKING\GIS\02_Workspace

06.2

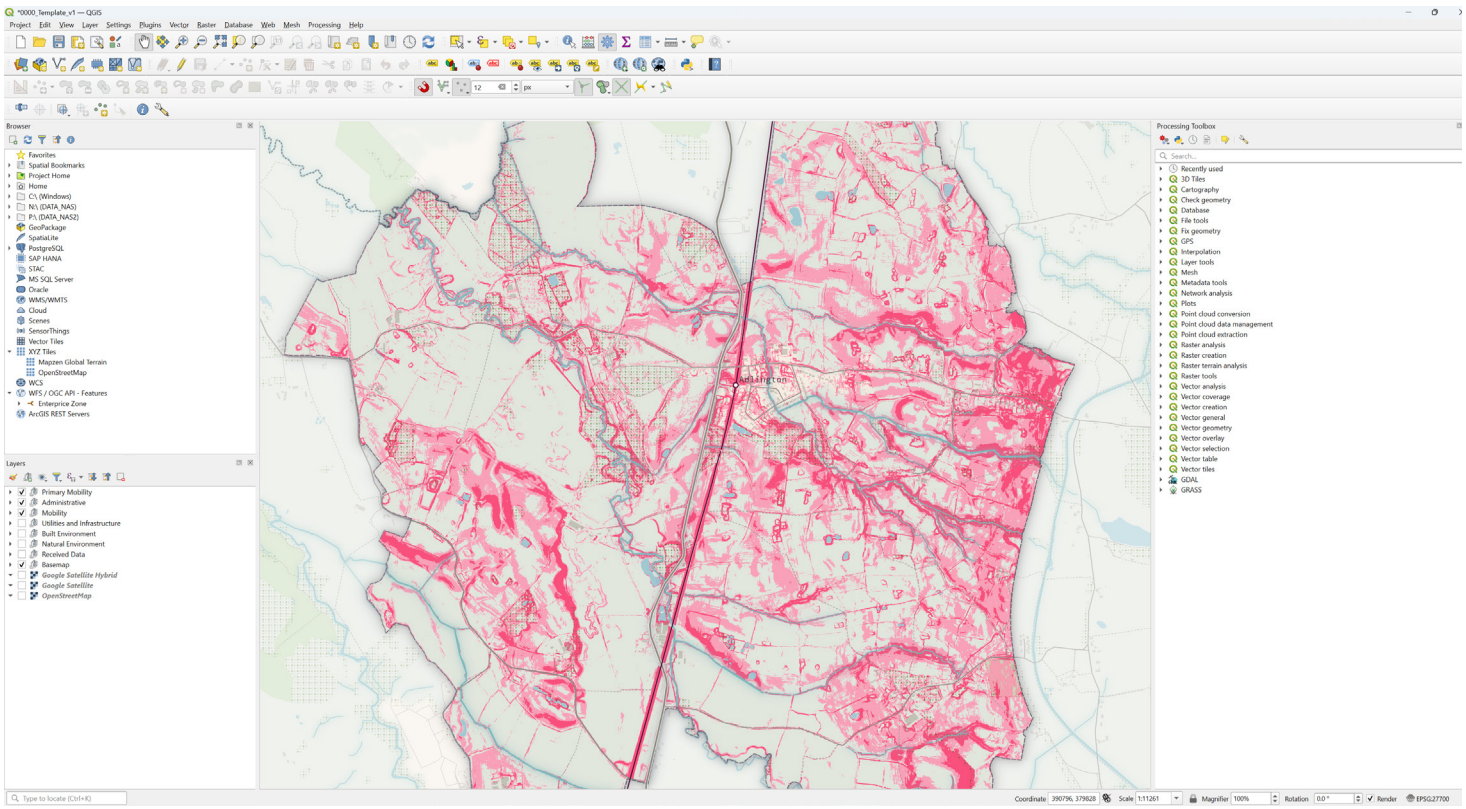
How to open the workspace

01 Navigate to your project's GIS folder on the server:
N:\00_XX_PROJECT\WORKING\GIS\02_Workspace

02 Double-click the .qgz file. QGIS will open with all pre-loaded layers and the correct database connection.

03 In the Layers Panel (bottom-left), you will see the baseline layers organised by theme group (e.g. Mobility, Built Environment, Basemap).

04 The map canvas will be zoomed to your project area.



06.3

Understanding your access

Not all data in the workspace can be edited. This protects baseline datasets from accidental changes.

— Read-only access (baseline schema): You can view, query, and export these layers, but you cannot edit the data. These are the company-wide datasets maintained by the DI team. If you need a modified version of a baseline layer, copy it to your project schema first.

— Write access (your project schema): You can add, edit, and delete data in your assigned project schema. This is where project-specific datasets, client-provided data, and your own analysis outputs should be stored.

— How to tell the difference: In the Layers Panel, right-click a layer and select Toggle Editing. If the layer is read-only, QGIS will display a warning that the layer is not editable. Baseline layers from the database will always behave this way.

06.4

Key rules

— Never attempt to edit baseline layers directly. Always copy to your project schema first.
— Save your QGIS project file regularly, but do not rename or move it from the server location.
— If you need additional baseline datasets not already in the workspace, contact the DI team.
— If you need a new project schema or additional write access, contact Abhi, Po, or Sandra.

07
Importing & exploring data

This section covers how to browse the geodatabase, add layers to your workspace, and explore the data they contain.

07.1

Browsing schemas and locating layers

01 In the Browser Panel, expand PostGIS to see all available schemas.

02 Expand a schema to see its layers. For most projects, start with uk_baseline which contains all available datasets.

03 Layers are named by theme, source, and content. Use the theme code at the start of each layer name to find what you need:

→ hth_ = Health & Wellbeing
→ mob_ = Transport & Mobility
→ adm_ = Administrative Boundaries
→ env_ = Environment & Natural Assets

Refer to p.19 for the full theme code table.

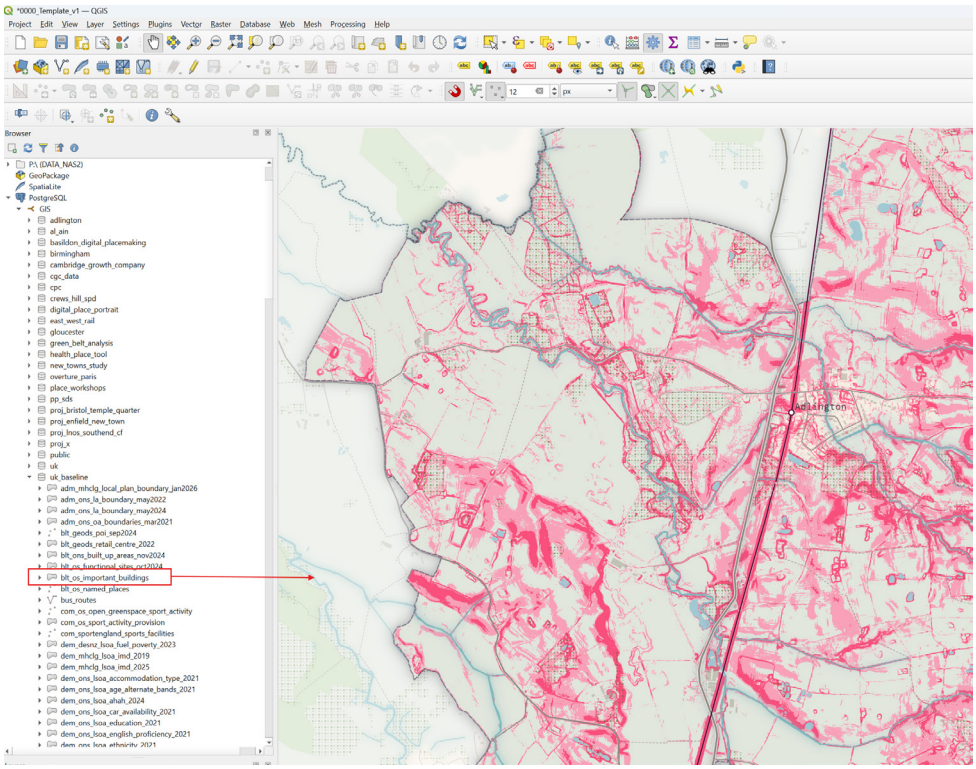
07.2

Adding layers to your workspace

01 Locate the layer you want in the Browser Panel.

02 Double-click the layer, or drag it directly into the map canvas or the Layers Panel.

03 The layer will appear in your Layers Panel and render on the map canvas.



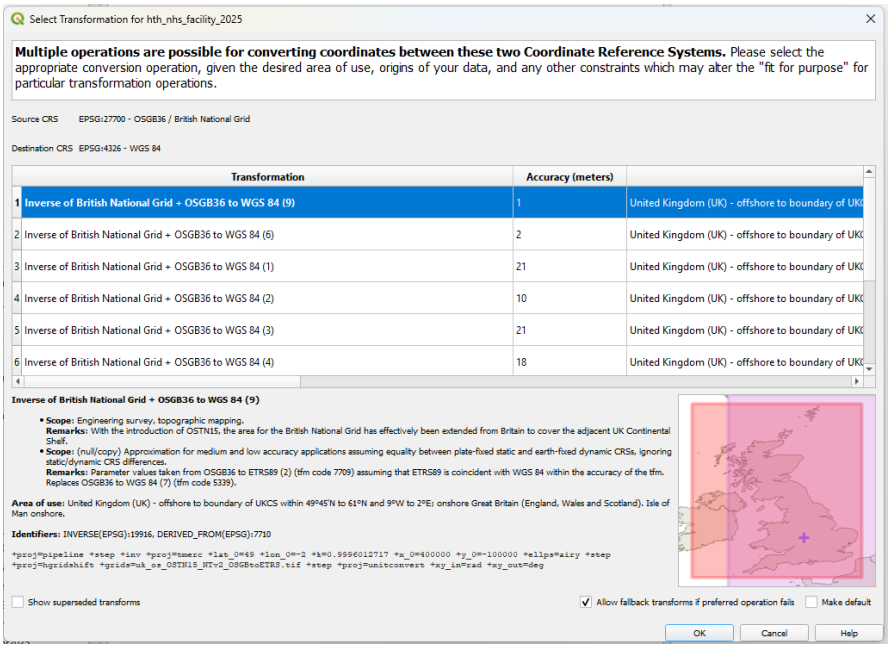
07.3

Handling CRS prompts

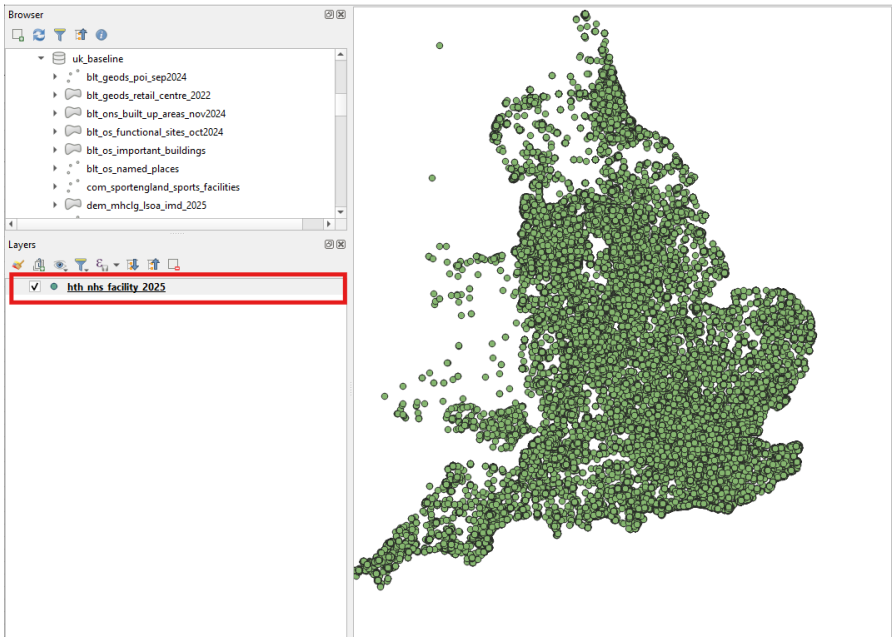
When you add a layer from the geodatabase, QGIS may prompt you to select a coordinate transformation. This happens because baseline data is stored in British National Grid (EPSG:27700) and your project may use a different display CRS.

When prompted:
Select the first option — Inverse of British National Grid + OSGB36 to WGS 84 (9) then Click OK

This is the most accurate transformation for UK data. If you are unsure, accept the default selection — it will be correct for all standard UK projects.



Once successfully imported, the layer will appear in your workspace under the Layers Panel.



07.4

Viewing attribute data

To explore the data behind a layer:

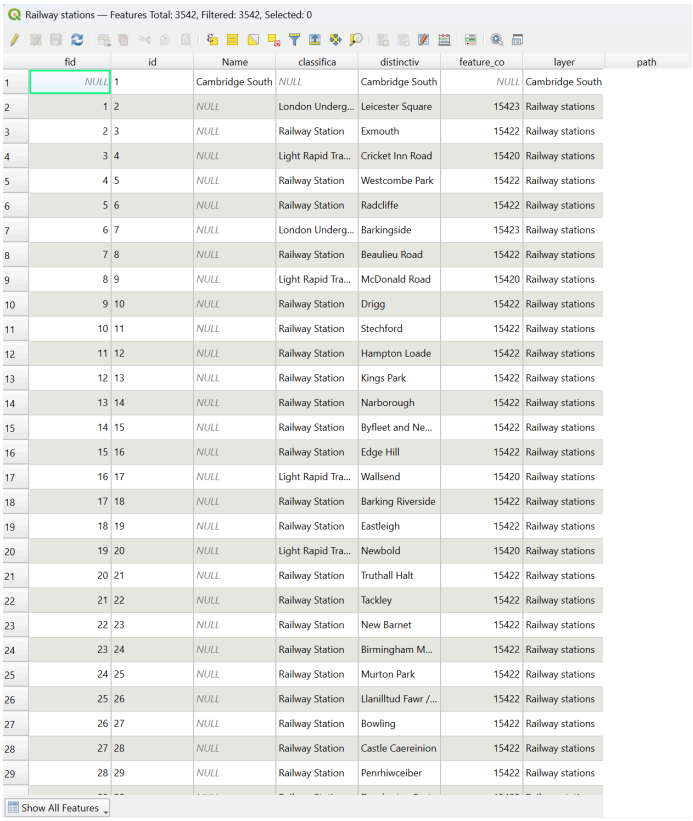
01 Right-click the layer in the Layers Panel (Not Browser Panel).

02 Select Open Attribute Table.

03 The attribute table displays all fields and records for that layer. You can sort, filter, and search within this table.

Useful actions in the attribute table:
— Click a column header to sort by that field
— Click the filter icon (funnel) to filter records by value
— Select a row to highlight that feature on the map
— Use the expression filter button to write custom queries (e.g. la_name = ‘Camden’)

To identify a single feature on the map without opening the full table:
— Click the Identify tool (blue circle with white “i” icon) in the toolbar
— Click any feature on the map
— A panel will show all attribute values for that feature



	fid	id	Name	classifica	distinctiv	feature_co	layer	path
1	NULL	1	Cambridge South	NULL	Cambridge South	NULL	Cambridge South	
2	1	2	NULL	London Underg...	Leicester Square	15423	Railway stations	
3	2	3	NULL	Railway Station	Exmouth	15422	Railway stations	
4	3	4	NULL	Light Rapid Tra...	Cricket Inn Road	15420	Railway stations	
5	4	5	NULL	Railway Station	Westcombe Park	15422	Railway stations	
6	5	6	NULL	Railway Station	Radcliffe	15422	Railway stations	
7	6	7	NULL	London Underg...	Barkingside	15423	Railway stations	
8	7	8	NULL	Railway Station	Beaulieu Road	15422	Railway stations	
9	8	9	NULL	Light Rapid Tra...	McDonald Road	15420	Railway stations	
10	9	10	NULL	Railway Station	Drigg	15422	Railway stations	
11	10	11	NULL	Railway Station	Stechford	15422	Railway stations	
12	11	12	NULL	Railway Station	Hampton Loade	15422	Railway stations	
13	12	13	NULL	Railway Station	Kings Park	15422	Railway stations	
14	13	14	NULL	Railway Station	Narborough	15422	Railway stations	
15	14	15	NULL	Railway Station	Byfleet and Ne...	15422	Railway stations	
16	15	16	NULL	Railway Station	Edge Hill	15422	Railway stations	
17	16	17	NULL	Light Rapid Tra...	Wallsend	15420	Railway stations	
18	17	18	NULL	Railway Station	Barking Riverside	15422	Railway stations	
19	18	19	NULL	Railway Station	Eastleigh	15422	Railway stations	
20	19	20	NULL	Light Rapid Tra...	Newbold	15420	Railway stations	
21	20	21	NULL	Railway Station	Truthall Halt	15422	Railway stations	
22	21	22	NULL	Railway Station	Tackley	15422	Railway stations	
23	22	23	NULL	Railway Station	New Barnet	15422	Railway stations	
24	23	24	NULL	Railway Station	Birmingham M...	15422	Railway stations	
25	24	25	NULL	Railway Station	Murton Park	15422	Railway stations	
26	25	26	NULL	Railway Station	Llanilltud Fawr /...	15422	Railway stations	
27	26	27	NULL	Railway Station	Bowling	15422	Railway stations	
28	27	28	NULL	Railway Station	Castle Caereinion	15422	Railway stations	
29	28	29	NULL	Railway Station	Penrhiwceiber	15422	Railway stations	

08
Data import & storage

When you receive or download new data, it needs to be saved in the correct location before you start working with it. This section covers where to store data and what to check before using it.

08.1

Where to save data

All incoming data (from clients, partners, or public sources) must be saved to the project server immediately, not on your local machine.

— Client-provided or externally sourced data:
N:\00_XX_PROJECT\Source IN\From_[SenderName]\YYMMDD_File_Name

— Data you create during project work:
N:\00_XX_PROJECT\WORKING\GIS\01_Data

Follow the naming standards in section 09 when saving your files.

08.2

Dataset verification

— Licence: Are we permitted to use this data for our intended purpose?
— Geographic coverage: Does it cover the project area?
— Format: What is the source format? (Shapefile, GeoPackage, GeoJSON, CSV, FileGDB)
— CRS: What coordinate reference system does it use? UK data should be EPSG:27700

If the licence or CRS is unclear, check with the DI team before proceeding.

08.3

Standard workflow

01 Save data to the correct server location

02 Open your pre-configured QGIS workspace (section 06)

03 Import datasets from the geodatabase or server (section 07)

04 Work — analyse, visualise, and save outputs to GIS\01_Data

09
Naming standards

When saving data to the project server, follow these conventions to keep files organised and searchable.

09.1
General rules

— Lowercase only
— Do not start names with a number
— Avoid special symbols, no dots, commas, quotes, slashes, semicolons
— Use underscores to separate words with _ (not spaces or hyphens)
— Be descriptive but concise

09.2
Server file naming

<project number>_<project name>_<theme>_<source>_<layername>_<existing or proposed>_v<version>.<ext>

Example: 1938_regis_road_kentish_town_mob_osm_railway_existing_v02.gpkg

Refer to section 04 for theme codes (ADM, BLT, MOB, etc.)

Database table naming is managed by the DI team and covered in the Office Data Handling Protocols.

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